**UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS**

**GCE Ordinary Level**

# MARK SCHEME for the May/June 2008 question paper

## 5090 BIOLOGY

**5090/06** Paper 6 (Alternative to Practical), maximum raw mark 40

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**1 (a)**

| test–tube / no. | total surface area / $cm^2$ | time taken for colour change / s [/ min] |
|---|---|---|
| 1 | 6 | 510 [8.5] |
| 2 | 12 | 45 [0.75] |
| 3 | 8 | 225 [3.75] |

marks: 1 format as shown **A**: horizontal or vertical and extra column(s)
2 table ruled and joined up
3 titles and units as appropriate in headers
4 surface area calculations correct
5 all boxes completed **R**: mixed minutes and seconds [5]

**(b) (i)** graph marks: [5]
1 axes correct (*x* – surface area / volume – horizontal)
2 labelled 'surface area / volume (ratio)' and 'time / seconds' (t / s)
3 correct (equal spaced) scale, good size
4 plots **clear** and accurate
5 good line of best fit / ruled connections
bar: 1 and 2 only

**(ii)** bigger surface area (volume ratio) – shorter time / faster diffusion rate ; [1]
**A**: inversely proportional

**(c)** determination of end point ; **R**: 'timing'
accuracy of block size ;
effect on surface area of blocks clumping ;
ovp ; e.g. ensure blocks covered by **A2** [up to 2]

**(d)** living cell has (cell) membrane ;
materials moved through (semi-permeable ) membrane / active transport ; **R**: osmosis
cytoplasm (of uneven density) ;
uneven / variable shape ; (c.f. cube)
ovp ; [up to 2]

**(e)** apparatus assembled correctly (diagram to include thermometer) ;
same size / surface area blocks of agar ;
different (static) temperatures ;
range of temperatures suggested (2 will do, **R**: boiling) ;
record results /data / plot graphs ;
same volume of **A2** ;
replication / repeats / mean values ;
ovp ; e.g. temperature constant before blocks put in [up to 6]

**[Total: 21]**

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**2** **(a)** **(i)** drawing marks: [D.2]
1 clear, clean lobes, at least 7 cm
2 variegation shown (not just shaded)

labels: 2 correct from –
green and white / yellow areas / variegated chlorophyll absent / present ;
vein ;
lamina ; [up to 2]

**(ii)** width of **L1** 67–68 mm ;
line drawn and correctly measured, units correct once ; [2]

**(iii)** working expression correct ; **A**: stated in words
magnification correct and well expressed ; [2]
up to 2 d.p., not more than 0.2 rounding up/down

**(b)** **(i)** increases permeability / denatures enzymes / stops reactions /
/ kills cells/leaf ; **R**: kill enzymes [1]

**(ii)** remove chlorophyll / decolourise ; **R**: chloroplasts [1]

**(iii)** test for starch ; [1]

**(c)** starch produced where chlorophyll present ;
white area produces no starch ; **A**: converse
chlorophyll harnesses light / energy (for photosynthesis) ;
another detail – e.g. need to decolourise ; [up to 3]

**(d)** drawing marks: [D.3]
1 complete section, at least 7 cm deep, clear and realistic
2 upper cuticle ± correct
3 stoma shown – 2 guard cells correct

labels:
2 from: stoma(ta), guard cell, epidermis, cuticle ; [1]
2 from: palisade, spongy, air (intercellular) space, mesophyll ; [1]

**[Total: 19]**

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